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Highlights:

— An overview of the unprecedented hydrological event that occurred in the state of Rio Grande do Sul.
— Record floods ran from small headwater rivers to the Patos Lagoon, which drains more than 180,000 km^2.
— The April–May hydrological event offers a glimpse of what may be expected in the coming decades due to climate change.

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Correspondence to:

Walter Collischonn
collischonn@iph.ufrgs.br

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The authors declare no competing interests.

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The exceptional hydrological disaster of April–May 2024 in southern Brazil

Walter Collischonn[1,*], Fernando Mainardi Fan[1], Iporã Possantti[1], Fernando Dornelles[1], Rodrigo Paiva[1], Matheus Sampaio Medeiros[1], Gean Paulo Michel[1], Fernando Jorge Corrêa Magalhães Filho[1], Sofia Royer Moraes[2], Francisco Fernando Noronha Marcuzzo[3], Rossano Dalla Lana Michel[1], Tamara Leitzke Caldeira Beskow[4], Samuel Beskow[4], Elisa Helena Leão Fernandes[5], Leonardo Laipelt dos Santos[1], Anderson Ruhoff[1], Masato Kobiyama[1], Gilberto Loguercio Collares[4], Franco Buffon[3], Emanuel Duarte[3], Stefany Lima[1], Fernando Setembrino Cruz Meirelles[1], Daniel Gustavo Allasia Piccilli[6]

[1]Universidade Federal do Rio Grande do Sul, Porto Alegre, RS, Brazil; [2]Universidade do Vale do Taquari, Porto Alegre, RS, Brazil; [3]Serviço Geológico do Brasil, Porto Alegre, RS, Brazil; [4]Universidade Federal de Pelotas, Pelotas, RS, Brazil; [5]Universidade Federal do Rio Grande, Rio Grande, RS, Brazil; [6]Universidade Federal de Santa Maria, Santa Maria, RS, Brazil

Abstract We present an overview of the unprecedented hydrological event that occurred in Rio Grande do Sul state in late April and early May 2024, causing floods and landslides in a scale that was not seen before in Brazil. The rainfall that caused the disaster is incomparable to any recent event in Brazil at large spatial and temporal scale. The intense and long-lasting precipitation led to record floods from small rivers with drainage areas below 1,000 km^2 to the Patos Lagoon, which drains more than 180,000 km^2. The rainfall also led to landslides that rank first in Brazil in terms of spatial scale and number of occurrences. Mountain rivers experienced flash floods with extreme water levels, while in the lowlands the large rivers and the Patos Lagoon experienced long duration flooding, with stage remaining high for about 30 days. The flood event is probably ranked first in terms of displaced people and damage to property and infrastructure in Brazil. Almost 200 fatalities occurred due to the disaster and protection structure failures. The April–May hydrological event show a glimpse of what can be expected along the coming decades due to climate change in a region that covers South Brazil, and parts of the neighboring countries.

Keywords: Natural disaster; Extreme hydrological events; Exceptional floods.

Resumo Apresentamos uma visão geral do evento hidrológico sem precedentes que ocorreu no estado do Rio Grande do Sul no final de abril e início de maio de 2024, causando inundações e movimentos de massa em uma escala nunca vista no Brasil. A precipitação que causou o desastre é incomparável a qualquer evento recente no país em termos de escala espacial e temporal. A intensa e prolongada precipitação resultou em enchentes recordes, desde pequenos rios com áreas de drenagem abaixo de 1.000 km^2 até a Laguna dos Patos, que drena mais de 180.000 km^2. As chuvas também levaram a movimentos de massa que ocupam o primeiro lugar no Brasil em termos de extensão espacial e número de ocorrências. Enchentes rápidas com níveis extremos de água ocorreram em regiões montanhosas, enquanto nas áreas mais baixas as cheias tiveram longa duração, com os níveis de água permanecendo elevados por cerca de 30 dias. Este desastre provavelmente ocupa a primeira posição no Brasil em número de pessoas deslocadas e danos a propriedades e infraestruturas. Quase 200 fatalidades ocorreram devido ao desastre e falhas em estruturas de proteção. O evento hidrológico de abril e maio oferece um vislumbre do que pode ser esperado nas próximas décadas devido às mudanças climáticas em uma região que abrange o Sul do Brasil e partes dos países vizinhos.

Palavras-chave: Desastre natural; Eventos hidrológicos extremos; Enchentes excepcionais.

1 Introduction

Between the end of April and the beginning of May 2024, heavy rains caused extraordinary flooding in southern Brazil. The largest floods and landslides ever observed occurred along several rivers in the state of Rio Grande do Sul, such as the Jacuí, Taquari, Caí, Sinos and Guaíba (New York Times 2024). The Patos Lagoon, which ultimately receives the water of all those rivers, also reached the highest levels ever recorded. In the state's capital, Porto Alegre, the Guaíba flood exceeded the maximum level observed during the traumatic flood of 1941 (Possa et al. 2022; Silveira et al. 2023), with even more devastating consequences because the flood protection system, built in the 1970 decade, was unable to prevent the water from entering the city.

The hydrological disaster occurred in four forms: (1) mass movements in the mountains, (2) flash floods in rivers in the transition between the mountains and the plains, (3) rel-

atively slower and long-lasting flooding along the densely populated areas close to the lowland rivers, where the failure of the flood protection systems intensified the impacts, and (4) slow flooding throughout the margins of the Patos Lagoon.

The exceptional hydrological disaster of April–May 2024 in southern Brazil not only marks new records of flooding in several rivers in the region; it is also the most extensive landslide event ever observed in Brazil (Andrades Filho et al. 2024; Brasil. Centro Nacional de Monitoramento e Alertas de Desastres Naturais – CEMADEN 2024). Hundreds of thousands of people were directly affected by the floods, which inundated more than 15,000 km^2. Fatal victims are at least 183, and 27 people are still reported missing, five months after the disaster (Rio Grande do Sul 2024).

The 2024 extreme hydrological event occurred a few months after other noteworthy rainfall-related disasters in Brazil (Marengo, Alcantara, et al. 2023; Marengo, Cunha, et al. 2024; Marengo, Seluchi, et al. 2023), and only eight months after a catastrophic flood hit the same southern Brazilian region in September 2023 (Alvalá et al. 2024; Moraes et al. 2024), suggesting that the region is passing through a particular flood-rich period. This recent flood-rich period in southern Brazil seems to be in accordance with already reported trends in observed rainfall and floods during the last decades (Chagas et al. 2022) and with projections of possible hydrological impacts of climate change in southern Brazil for the twenty-first century (Brêda et al. 2023; Hodnebrog et al. 2022).

Given its profound, widespread and long-lasting impacts, the disaster of April–May 2024 can be considered the biggest natural hydrological disaster in the history of Brazil. It also gives an example of an event that can be called an "impossible flood", as suggested by Montanari et al. (2024), because it exceeded the expectations based on historical experience, and because it exposed the consequences of the occurrence of an event whose probability of occurrence was considered too small to act, even by the flood managers (Allasia et al. 2015).

The objective of this study is to document this exceptional hydrological event and the disaster that followed. We aimed to describe the process chain of the April–May 2024 hydrological disaster in South Brazil, from causes to impacts, focusing on what made the impacts so exceptional.

In our understanding, adequately documenting extraordinary hydrological events helps society learn from those events, by understanding patterns and causes. This learning process helps reduce vulnerability and increase resilience to future disasters (Zanandrea et al. 2023). Other examples of papers that describe extraordinary floods are given by Gaume et al. (2004), Dimitriou et al. (2024), Mohr et al. (2023), Nanditha et al. (2023) and Kreibich et al. (2017). In the Brazilian context, accounts of other extraordinary hydrological events have been recently organized by Zanandrea et al. (2023). We also understand that a detailed description and study of such an extraordinary flood event is not of exclusive interest to the affected region, because the lessons that can be drawn from it are not limited to the basin or river in which it occurred but can be extremely relevant to an entire large region, as suggested by Merz et al. (2024).

The paper is structured as follows: Section 2 describes the data sets and methods used. Section 3 describes the event in terms of meteorological and hydrological aspects and its impacts. Finally, Section 4 discusses and summarizes the main results and draws conclusions.

2 Methods

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2.1 Citing sources

References live in `references.bib` and are formatted in APA style with author–year citations. There are two forms, and the choice is grammatical rather than cosmetic.

Use `\parencite{key}` when the citation supports a statement but is not part of the sentence, so it belongs in parentheses: the Patos Lagoon is the largest choked lagoon in South America (Chagas et al. 2022). Use `\textcite{key}` when the authors *are* the subject of the sentence, so their names need to read as words: Montanari et al. (2024) calls such an event an “impossible flood”.

Both take several keys at once, separated by commas — `\parencite{a,b}` — and biblatex orders and punctuates the group for you (Marengo, Alcantara, et al. 2023; Marengo, Cunha, et al. 2024; Marengo, Seluchi, et al. 2023). Every citation is a clickable cross-reference to its entry in the reference list.

Cross-references to document elements use `\Cref{label}`, which supplies the word as well as the number (Section 1), and `\cref{label}` inside a sentence, as in section 2.4 below. Give every figure, table, equation and section a label and never type a number by hand.

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2.2 Mathematical features

Inline mathematics is written between dollar signs, as in the continuity relation $Q = vA$, where Q is discharge, v mean velocity and A the cross-sectional area. Keep short expressions inline; give anything you need to refer to a display of its own.

Numbered display equations go in an `equation` environment with a label, so they can be cross-referenced:

$$v = \frac{1}{n} R^{2/3} S^{1/2}. \tag{1}$$

Equation (1) is referenced by label, so its number stays correct however many equations are inserted before it. Use `\Cref` at the start of a sentence and `\cref` inside one.

An unnumbered display uses the starred form, for a step that needs room but never gets referred to:

$$R = \frac{A}{P}.$$

Related equations that belong together can share a number and be lettered individually with `subequations`:

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = q_\ell, \tag{2a}$$

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x}\left(\frac{Q^2}{A}\right) = gA\left(S_0 - S_f\right). \tag{2b}$$

Equation (2) is the pair; eq. (2a) and eq. (2b) are its parts, numbered **a** and **b** automatically.

Systems are aligned on a chosen character with `align`, one `&` per alignment point:

$$\begin{aligned} S_f &= \frac{n^2 v^2}{R^{4/3}}, & F_r &= \frac{v}{\sqrt{g h}}, \\ \tau_0 &= \rho\, g R S_f, & \mathrm{Re} &= \frac{v R}{\nu}. \end{aligned} \tag{3}$$

Symbols take the usual variants: Greek letters α, β, λ, μ, σ, Δ, Ω; decorated forms $\bar{x}$, $\hat{y}$, $\tilde{z}$, $\dot{Q}$, $\vec{u}$; calligraphic $\mathscr{R}$, blackboard $\mathbb{R}$, bold $\mathbf{A}$ and roman subscripts such as Q_{peak}. Operators and relations behave as expected: $\sum$, $\prod$, $\int$, ∂, ∇, $\leq$, $\approx$, $\propto$, $\in$.

Theorem-like environments are available for formal statements, and number independently of the equations:

Definition 1 (Specific energy)**.** *For a channel of depth h carrying mean velocity v, the specific energy is $E = h + v^2/2g$.*

Theorem 1 (Critical flow)**.** *Specific energy is minimised when $F_r = 1$.*

Proof. Differentiate E from definition 1 with respect to h at fixed discharge and set $\mathrm{d}E/\mathrm{d}h = 0$; the result rearranges to $v^2 = g\,h$, which is $F_r = 1$ by eq. (3). □

Remark. *`lemma`, `corollary`, `example` and `remark` are defined alongside these, and all follow the document language.*

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2.3 Developing figures

Figures come in three sizes, each a single macro. Give the image file's bare name — no folder, no extension — and the template looks it up in `figures/`, preferring `.pdf`, then `.png`, then `.jpeg`, then `.jpg`:

- `\FigureM{name}{caption}{label}` — text width, as in Figure 1.
- `\FigureS{name}{caption}{label}` — small, with the caption set beside the image and spilling into the sidebar column, as in Figure 2. Pass a second optional argument to change its width, for example `\FigureS[t][0.5]{...}`; the default is 0.67 of the text width.
- `\FigureL{name}{caption}{label}` — full width, margin to margin, as in Figure 3.

Every one of them takes an optional float position first, so `\FigureM[b]{...}` sends that figure to the bottom of a page. The default is `t`, the top.

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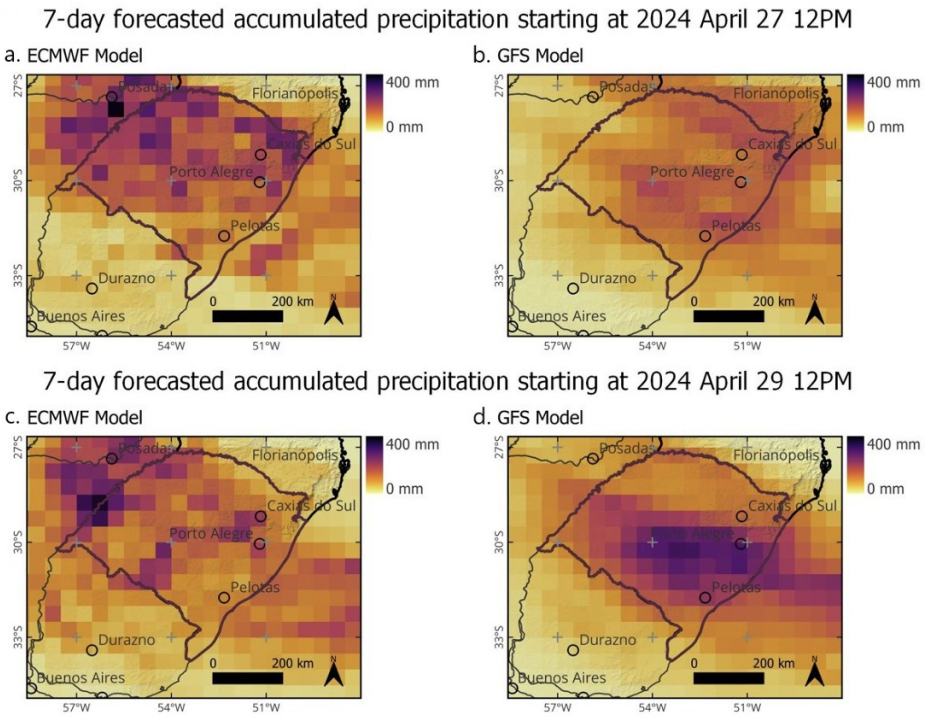


Figure 1 Seven-day accumulated precipitation forecasts from the ECMWF and GFS global models for the river basins of Rio Grande do Sul. Darker shading indicates higher forecast accumulation; both models anticipated the heavy rainfall that preceded the flood. **a** — ECMWF, forecast issued 27 April. **b** — GFS, forecast issued 27 April. **c** — ECMWF, forecast issued 29 April. **d** — GFS, forecast issued 29 April.

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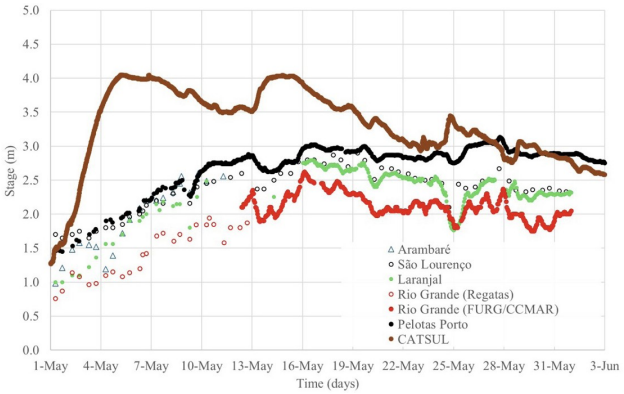


Figure 2 River stage (water level) from 1 May to 3 June 2024 at six gauging stations across the Rio Grande do Sul lowlands and coastal lagoon system (Arambaré, São Lourenço, Laranjal, Rio Grande, Pelotas Porto and CATSUL), showing the sustained high-water period that followed the flood event.

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Figure 3 Aerial and ground documentation of slope failures and structural damage following the April–May 2024 event. **a** — debris-flow scars draining into a stream channel. **b** — hillslope cracking upstream of a residential area. **c** — a large landslide scar above a rural property. **d** — structural cracking in a house wall. **e** — a building destroyed by a landslide.

ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

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If an image file cannot be found, the build still succeeds and a placeholder appears in its place, naming the file it looked for — see Figure 4. A missing or misspelled image is therefore visible in the PDF instead of stopping the compile. Delete this example from your own paper.

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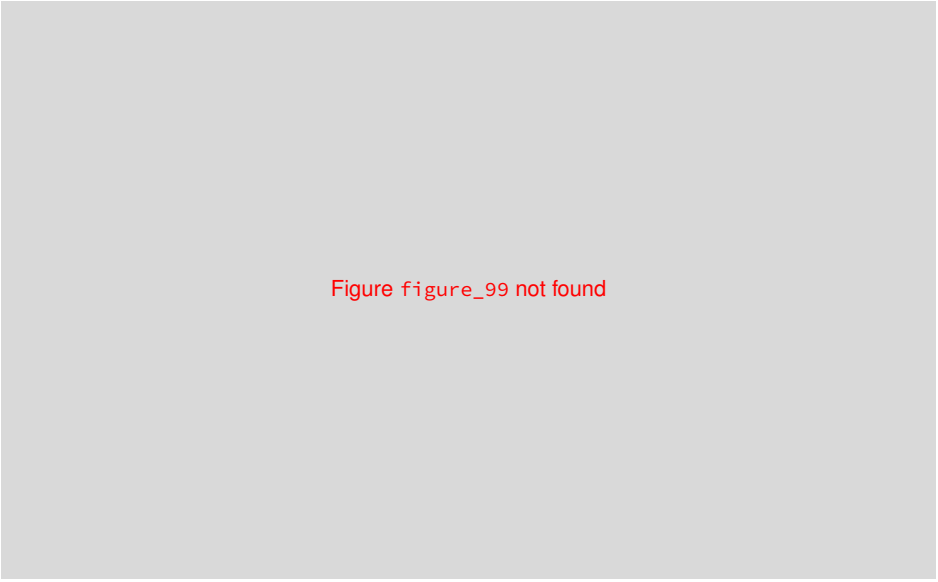


Figure 4 Placeholder shown when an image file cannot be found in `figures/`. The box keeps the golden ratio of the requested width.

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2.4 Presenting tables

Tables come in two sizes and are written as environments, so the tabular material goes in the body and the caption is supplied as an argument. The caption is placed above the table, following the usual convention:

- `\begin{TableM}{caption}{label}` — text width, as in Table 1.
- `\begin{TableL}{caption}{label}` — full width, margin to margin, for tables with many columns, as in Table 2.

Both accept the same optional position argument as the figure macros, defaulting to `t`. Use `booktabs` rules (`\toprule`, `\midrule`, `\bottomrule`) and avoid vertical lines. Footnotes to a table go in a `tablenotes` block at the end of the environment, keyed with `\tnote{a}` in the cell they annotate.

Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur

Table 1 | Gauging stations used in this study, with drainage area and the peak stage recorded during the event.

Station	River	Area (km^2)	Peak (m)
Arambaré	Patos Lagoon	180,000	2.6[a]
São Lourenço	Patos Lagoon	180,000	2.8
Laranjal	Patos Lagoon	180,000	2.8
Pelotas Porto	São Gonçalo	4,800	3.0

[a] Notes go in a `tablenotes` block and wrap to the width of the table above them.

ridiculus mus. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetuer eu, nonummy id, sapien. Nullam at lectus. In sagittis ultrices mauris. Curabitur malesuada erat sit amet massa. Fusce blandit. Aliquam erat volutpat. Aliquam euismod. Aenean vel lectus. Nunc imperdiet justo nec dolor.

Etiam euismod. Fusce facilisis lacinia dui. Suspendisse potenti. In mi erat, cursus id, nonummy sed, ullamcorper eget, sapien. Praesent pretium, magna in eleifend egestas, pede pede pretium lorem, quis consectetuer tortor sapien facilisis magna. Mauris quis magna varius nulla scelerisque imperdiet. Aliquam non quam. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl.

The large variant spans the full width, which makes room for tables that would otherwise be cramped in the text column.

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3 Results

Nulla in ipsum. Praesent eros nulla, congue vitae, euismod ut, commodo a, wisi. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Aenean nonummy magna non leo. Sed felis erat, ullamcorper in, dictum non, ultricies ut, lectus. Proin vel arcu a odio lobortis euismod. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Proin ut est. Aliquam odio. Pellentesque massa turpis, cursus eu, euismod nec, tempor congue, nulla. Duis viverra gravida mauris. Cras tincidunt. Curabitur eros ligula, varius ut, pulvinar in, cursus faucibus, augue.

Nulla mattis luctus nulla. Duis commodo velit at leo. Aliquam vulputate magna et leo. Nam vestibulum ullamcorper leo. Vestibulum condimentum rutrum mauris. Donec id mauris. Morbi molestie justo et pede. Vivamus eget turpis sed nisl cursus tempor. Curabitur mollis sapien condimentum nunc. In wisi nisl, malesuada at, dignissim sit amet, lobortis in, odio. Aenean consequat arcu a ante. Pellentesque porta elit sit amet orci. Etiam at turpis nec elit ultricies imperdiet. Nulla facilisi. In hac habitasse platea dictumst. Suspendisse viverra aliquam risus. Nullam pede justo, molestie nonummy, scelerisque eu, facilisis vel, arcu.

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Table 2 | Summary of the sub-basins affected by the April–May 2024 event, with rainfall totals, peak discharge and the recurrence interval estimated for each.

Sub-basin	Area (km^2)	Rainfall (mm)	Peak (m^3/s)	Date	Return period (years)
Taquari	26,500	520	14,000	2024-05-02	>100
Caí	4,900	480	5,200	2024-05-01	>100
Sinos	3,700	460	4,100	2024-05-03	>100
Jacuí	71,600	410	19,500	2024-05-04	>100[a]
Gravataí	2,000	390	980	2024-05-05	50
Taquari	26,500	520	14,000	2024-05-02	>100
Caí	4,900	480	5,200	2024-05-01	>100
Sinos	3,700	460	4,100	2024-05-03	>100
Jacuí	71,600	410	19,500	2024-05-04	>100[a]
Gravataí	2,000	390	980	2024-05-05	50
Taquari	26,500	520	14,000	2024-05-02	>100
Caí	4,900	480	5,200	2024-05-01	>100
Sinos	3,700	460	4,100	2024-05-03	>100
Jacuí	71,600	410	19,500	2024-05-04	>100[a]
Gravataí	2,000	390	980	2024-05-05	50

[a] Return periods above 100 years are reported as a lower bound, the record being too short to resolve them.

Praesent scelerisque. Maecenas posuere sodales odio. Vivamus metus lacus, varius quis, imperdiet quis, rhoncus a, turpis. Etiam ligula arcu, elementum a, venenatis quis, sollicitudin sed, metus. Donec nunc pede, tincidunt in, venenatis vitae, faucibus vel, nibh. Pellentesque wisi. Nullam malesuada. Morbi ut tellus ut pede tincidunt porta. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam congue neque id dolor.

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4 Conclusions

Nulla non mauris vitae wisi posuere convallis. Sed eu nulla nec eros scelerisque pharetra. Nullam varius. Etiam dignissim elementum metus. Vestibulum faucibus, metus sit amet mattis rhoncus, sapien dui laoreet odio, nec ultricies nibh augue a enim. Fusce in ligula. Quisque at magna et nulla commodo consequat. Proin accumsan imperdiet sem. Nunc porta. Donec feugiat mi at justo. Phasellus facilisis ipsum quis ante. In ac elit eget ipsum pharetra faucibus. Maecenas viverra nulla in massa.

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Donec et nisl id sapien blandit mattis. Aenean dictum odio sit amet risus. Morbi purus. Nulla a est sit amet purus venenatis iaculis. Vivamus viverra purus vel magna. Donec in justo sed odio malesuada dapibus. Nunc ultrices aliquam nunc. Vivamus facilisis pellentesque velit. Nulla nunc velit, vulputate dapibus, vulputate id, mattis ac, justo. Nam mattis elit dapibus purus. Quisque enim risus, congue non, elementum ut, mattis quis, sem. Quisque elit.

Appendices

A Rain gauge dataset

Describe the first dataset here: where it came from, how it was processed, and any caveats a reader would need in order to reuse it.

Etiam pede massa, dapibus vitae, rhoncus in, placerat posuere, odio. Vestibulum luctus commodo lacus. Morbi lacus dui, tempor sed, euismod eget, condimentum at, tortor. Phasellus aliquet odio ac lacus tempor faucibus. Praesent sed sem. Praesent iaculis. Cras rhoncus tellus sed justo ullamcorper sagittis. Donec quis orci. Sed ut tortor quis tellus euismod tincidunt. Suspendisse congue nisl eu elit. Aliquam tortor diam, tempus id, tristique eget, sodales vel, nulla. Praesent tellus mi, condimentum sed, viverra at, consectetuer quis, lectus. In auctor vehicula orci. Sed pede sapien, euismod in, suscipit in, pharetra placerat, metus. Vivamus commodo dui non odio. Donec et felis.

Table A.1 | Structure of the rain gauge dataset: one record per gauge and day.

Field	Unit	Description
`gauge_id`	—	Station identifier
`date`	—	Observation date, ISO 8601
`rainfall`	mm	Daily accumulated rainfall
`quality`	—	Quality flag, 0–3

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B Streamflow dataset

Describe the second dataset here, following the same pattern. Figure B.1 shows that figures and tables inside an appendix pick up lettered numbering automatically.

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Maecenas non massa. Vestibulum pharetra nulla at lorem. Duis quis quam id lacus dapibus interdum. Nulla lorem. Donec ut ante quis dolor bibendum condimentum. Etiam egestas tortor vitae lacus. Praesent cursus. Mauris bibendum pede at elit. Morbi et felis a lectus interdum facilisis. Sed suscipit gravida turpis. Nulla at lectus. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Praesent nonummy luctus nibh. Proin turpis nunc, congue eu, egestas ut, fringilla at, tellus. In hac habitasse platea dictumst.

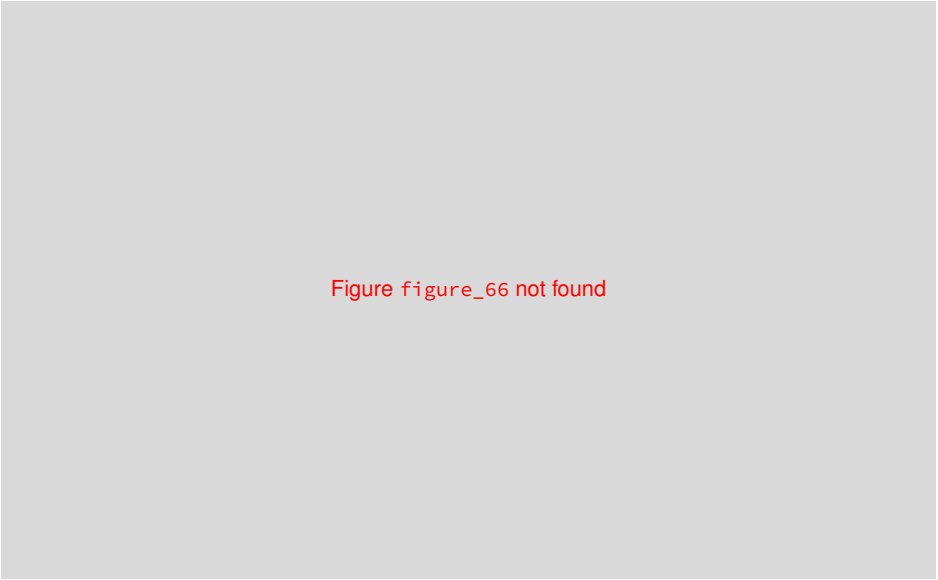


Figure B.1 Supplementary precipitation forecasts for the April–May 2024 event. See Figure 1 for model and date details.

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